

Ian Waudby-Smith

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Academic Positions

University of California, Berkeley
Miller Postdoctoral Fellow
Host: Michael I. Jordan

Berkeley, CA
2024–2027

Education

Carnegie Mellon University
PhD, Statistics
Thesis: “Advances in anytime-valid inference”
Advisor: Aaditya Ramdas
[Umesh K. Gavaskar Best Dissertation Award](#)

Pittsburgh, PA
2019–2024

University of Waterloo
BMath, Pure Mathematics & Statistics
5-year co-op program
[Dean's Honours List](#)

Waterloo, Ontario
2013–2018

Publications

- [1] Chase Mathis and **Ian Waudby-Smith**. Confidence horizons. *Under review at the Journal of the Royal Statistical Society, Series B*.
- [2] Keegan Harris, Brian W. Lee, **Ian Waudby-Smith**, Philip Amortila, Nika Haghtalab, and Michael I. Jordan. Post-training at the edge of detectability: A game-theoretic approach to fine-tuning. *Under review at the International Conference on Learning Representations (ICLR)*, 2026.
- [3] [\(αβ\)](#) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Gaffke’s confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *arXiv preprint arXiv:2607.18661*.
- [4] [\(αβ\)](#) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Combining e-values via demi-supermartingales. *Under review at the Annals of Applied Probability*.
- [5] Ricardo J. Sandoval*, **Ian Waudby-Smith***, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925*.
- [6] Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *Under review at the Journal of the Royal Statistical Society, Series B*, 2026.
- [7] Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments.

Major revision at Biometrika.

- [8] **Ian Waudby-Smith**, Ricardo J. Sandoval, and Michael I. Jordan. Universal log-optimality for general classes of e-processes and sequential hypothesis tests. *R&R at the Annals of Statistics*.
- [9] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *Major revision at Probability Theory and Related Fields*.
- [10] **Ian Waudby-Smith**, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform large sample anytime-valid inference and the Robbins–Siegmund distributions. *Major revision at Biometrika*.
- [11] $(\alpha\beta)$ Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *Bulletin of the London Mathematical Society (to appear)*, 2026.
- [12] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability (to appear)*, 2026.
- [13] **Ian Waudby-Smith**, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.
- [14] **Ian Waudby-Smith**, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/JMS Journal of Data Science (Invited contribution)*, 1(3):1–42, 2024.
- [15] **Ian Waudby-Smith** and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology (Discussion paper)*, 86(1):1–27, 2024.
- [16] **Ian Waudby-Smith**, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning (Oral presentation)*, 2023.
- [17] Akash V. Maharaaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.
- [18] **Ian Waudby-Smith**, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting (Best paper award)*, pages 124–139. Springer, 2021.
- [19] **Ian Waudby-Smith** and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems (Spotlight paper)*, 33:20204–20214, 2020.
- [20] **Ian Waudby-Smith**, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.
- [21] **Ian Waudby-Smith**, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

$(\alpha\beta)$ Alphabetical ordering.

*Co-lead author.

Teaching Experience

Carnegie Mellon University

Graduate Teaching Assistant

Pittsburgh, PA

2019–22

- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

Other Experience

Google Research

Student Researcher

Mentors: Jean Pouget-Abadie & Jennifer Brennan

New York, NY

Jun–Aug 2023

Microsoft Research

Research Intern

Mentor: Paul Mineiro

Redmond, WA

May–Aug 2022

Adobe Research

Research Intern

Mentors: David Arbour & Ritwik Sinha

San Jose, CA

Jun–Aug 2020

The Hospital for Sick Children (SickKids)

Research Student

Mentor: Eleanor Pullenayegum

Toronto, Ontario

Apr–Aug 2019

Health Data Science Lab, University of Waterloo

Research Assistant

Mentors: Joel Dubin & Joon Lee

Waterloo, Ontario

2016–18

Department of Statistics, University of Waterloo

Research Assistant

Mentor: Pengfei Li

Waterloo, Ontario

Apr–Aug 2017

Department of Pure Mathematics, University of Waterloo

Research Assistant

Mentor: Yu-Ru Liu

Waterloo, Ontario

Jan - Apr 2017

Cancer Care Ontario

Student Analyst

Mentor: Zhihui (Amy) Liu

Toronto, Ontario

Jan–Apr 2016

Professional Service

Thesis committee member

- Tyron Lardy (PhD in Statistics, Leiden University)
- Noah Liniger (MSc in Data Science, ETH Zurich)

Reviewer

- Annals of Statistics
- Bernoulli
- Biometrika
- Electronic Journal of Probability
- Electronic Journal of Statistics
- Journal of the American Statistical Association
- Journal of the Royal Statistical Society, Series B
- Management Science
- New England Journal of Data Science
- Sankhya A.
- Symposium on Theory of Computing (STOC)

University of California, Berkeley**Berkeley, CA***Volunteer*

- Miller symposium organizing committee
- Incoming Miller fellow initiative

Carnegie Mellon University**Pittsburgh, PA***Volunteer*

- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

Awards

Institute of Mathematical Statistics**Berkeley, CA***New Researcher Travel Award**2026***Miller Institute for Basic Research in Science****Berkeley, CA***Miller fellowship**2024-'27***Carnegie Mellon University Department of Statistics and Data Science****Pittsburgh, PA***Umesh K. Gavaskar Best Dissertation Award**2024***Statistical Society of Canada****St. John's, NL***Probability Section Student Research Presentation Award**2024***Amazon Science****Pittsburgh, PA***Graduate Research Fellowship**2023***University of Waterloo****Waterloo, Ontario***Waterloo Statistics Student Conference Presentation Award**2022***Carnegie Mellon University Department of Statistics and Data Science****Pittsburgh, PA***Teaching Assistant of the Year**2021*

Adobe Research <i>PhD Research Gift</i>	Pittsburgh, PA 2020
University of Waterloo <i>David Johnston International Experience Award</i>	Waterloo, Ontario 2018
The Natural Sciences and Engineering Research Council of Canada <i>NSERC Undergraduate Student Research Award</i>	Waterloo, Ontario 2017
University of Waterloo <i>President's Research Award</i>	Waterloo, Ontario 2016–17
University of Waterloo <i>University of Waterloo President's Scholarship of Distinction</i>	Waterloo, Ontario 2014

Presentations

Short courses

MBZUAI <i>A brief introduction to game-theoretic, safe, anytime-valid inference</i> Mini course consisting of 4 lectures	Abu Dhabi, UAE 2025
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ERC OCEAN retreat <i>A brief introduction to game-theoretic, safe, anytime-valid inference</i> Mini course consisting of 3 lectures	Venice, Italy 2024
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Department seminars

Department of Statistics at the London School of Economics <i>Log-optimality and multi-armed sequential hypothesis testing</i>	London, UK 2026
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Department of Statistics and Actuarial Science at the University of Waterloo <i>Log-optimality and multi-armed sequential hypothesis testing</i>	Waterloo, Ontario 2026
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Department of Statistics at University of California, Davis <i>Log-optimality and multi-armed sequential hypothesis testing</i>	Davis, CA 2026
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Department of Statistics at Iowa State University <i>Anytime-valid inference and uniform central limit theory</i>	Ames, IA 2026
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Division of Biostatistics at the University of California, Berkeley <i>Anytime-valid off-policy inference for contextual bandits</i>	Berkeley, CA 2025
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Department of Statistics at Stanford University <i>Anytime-valid inference and uniform central limit theory</i>	Stanford, CA 2025
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Statistics Seminar at the University of Copenhagen <i>Anytime-valid off-policy inference for contextual bandits</i>	Copenhagen, Denmark 2023
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Invited talks at conferences / workshops / research groups

Institute for Operations Research and the Management Sciences (INFORMS) <i>Multi-armed log-optimality and sequential testing</i>	San Francisco, CA 2026
Whendee L. Silver lab at the University of California, Berkeley <i>Anytime-valid inference for the environmental sciences</i>	Berkeley, CA 2026
Inria/Sierra Seminar <i>Combining e-values via demi-supermartingales</i>	Paris, France 2026
Cornell University Rising Stars in Statistics and Data Science Workshop <i>Advances in asymptotic anytime-valid inference</i>	Ithaca, NY 2026
Joint Statistical Meetings (JSM) <i>Asymptotic anytime-valid inference</i>	Boston, MA 2026
Adaptive Experimentation Workshop at University College London (Gatsby) <i>Log-optimality and multi-armed sequential hypothesis testing</i>	London, UK 2026
International Workshop on Sequential Methodologies <i>Asymptotic anytime-valid statistical inference</i> Recipient of the IMS New Researcher Travel Award	Washington, D.C. 2026
Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis) <i>On nonasymptotic confidence intervals for treatment effects in randomized experiments</i>	Stanford, CA 2026
National University of Singapore Young Mathematical Scientists Forum <i>Log-optimality of e-processes and sequential tests</i>	Singapore 2025
Stanford Data Driven Seminar (host: Stefan Wager) <i>Log-optimality of e-processes and sequential tests</i>	Stanford, CA 2025
MBZUAI-Berkeley Workshop <i>Anytime-valid off-policy inference for contextual bandits</i>	Abu Dhabi, UAE 2025
Inria/Sierra Seminar <i>Anytime-valid inference and uniform central limit theory</i>	Paris, France 2025
International Seminar on Selective Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Virtual 2024
CLIMB Workshop <i>Election audits via anytime-valid inference</i>	Berkeley, CA 2024
Workshop on Game-Theoretic Statistical Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Oberwolfach, Germany 2024
Fienberg Student Research Workshop at Carnegie Mellon University <i>Election audits via anytime-valid inference</i>	Pittsburgh, PA 2024

Joint Statistical Meetings <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
International Conference on Machine Learning <i>Extensions of randomized response for private confidence sets</i>	Honolulu, HI 2023
Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen) <i>Anytime-valid off-policy inference for contextual bandits</i>	Amsterdam, Netherlands 2023
University of Toronto / The Vector Institute (host: Dan Roy) <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
Copenhagen Causality Lab, University of Copenhagen <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Virtual 2023
Conference on Digital Experimentation (CODE@MIT) <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Cambridge, MA 2022
Microsoft Research Reinforcement Learning Discussion Group <i>Anytime-valid contextual bandit inference</i>	Virtual 2022
Computing and Mathematical Sciences, California Institute of Technology (host: Houman Owhadi) <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Microsoft Research <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Safe, Anytime-Valid Inference (SAVI) Workshop <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Eindhoven, Netherlands 2022
Carnegie Mellon University Computer Science Theory Lunch <i>Estimating means of bounded random variables by betting</i>	Pittsburgh, PA 2021
International Seminar on Distribution-Free Statistics <i>Estimating means of bounded random variables by betting</i>	Virtual 2021
E-Vote-ID: The International Conference for Electronic Voting <i>RiLACS: Risk-limiting audits via confidence sequences</i>	Virtual 2021
Spotify Experimentation Platform Team <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Vinted Science and Analytics Meetup <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
<i>Contributed conference talks</i>	
International Conference on Statistics and Data Science (ICSIDS) <i>Concentration inequalities for strong laws of large numbers</i>	Seville, Spain 2025
Statistical Society of Canada meeting <i>Distribution-uniform strong laws of large numbers</i> <i>Recipient of the Probability Section's Student Presentation Award</i>	St. John's, NL 2024

International Conference on Statistics and Data Science (ICSDS) <i>Distribution-uniform anytime-valid inference</i>	Lisbon, Portugal 2023
Waterloo Student Conference in Statistics, Actuarial Science, and Finance <i>Estimating means of bounded random variables by betting</i> Recipient of the Student Presentation Award	Waterloo, Ontario 2022
TPDP: Theory and Practice of Differential Privacy Workshop <i>Locally private nonparametric confidence intervals and sequences</i>	Baltimore, MD 2022
Statistical Society of Canada (SSC) Annual Meeting <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2022
NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Confidence sequences for sampling without replacement</i>	Virtual 2020
Statistical Society of Canada (SSC) Annual Meeting <i>Multi-state models for chronic kidney disease prevalence projections in Ontario</i>	St. Catharines, Ontario 2016